The Wigner transform is an L2 isometry up to the semiclassical factor

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doi:10.5802/jep.353; CC BY4.0

Editorial note. The counted statement is the author's labelled lemma 'L2normofWigner' (source lines 474-479); the proof environment begins on the next line after \end{lemma}, so the pairing is by adjacency.

Editorial note. The retained context is the author's own lead-in sentence, which states the role of the L2 relation for the Wigner transform.

Editorial note. No counter adjustment is retained: the publisher numbering is not reproduced and the item prints with the wrapper's own sequential numbering.

Editorial note. No retained line refers to a label, citation or figure outside the retained span, so the package declares no external reference; the retained preamble is the publisher layout adapter plus the native macro and theorem block of the same pinned source.

Editorial scope. One original result of the pinned publisher article, counted once: the L2 isometry identity for the Wigner transform (author label L2normofWigner).

Allowed standard prerequisites (editorial).

- Wigner transform. W[Q](x,xi) is the semiclassically scaled partial Fourier transform of Q in the relative variable.. The proof rewrites W[Q] as an inverse-scaled Fourier transform in the relative variable at source line 483.
- Plancherel's theorem. A Fourier transform preserves the L2 norm.. It is the step that converts the L2 norm of the displayed Fourier transform into the L2 norm of Q (source line 484).
- Semiclassical scaling. The prefactor (2 pi hbar)^{-d/2} converts the quantum L2 norm into the classical one.. It appears both in the statement and in the displayed rewriting of W[Q] at source line 483.

The L^2 norm of $W[Q]$ is related to the $\mathcal{L}^2$ norm of Q via the following relation:

Lemma 0.1. *Suppose $Q \in \mathcal{L}^2(\mathbb{R}^d;\mathbb{C})$. Then the Wigner transform of Q satisfies*

$$\|W[Q](x,\xi)\|_{L^2_{x,\xi}} = (2\pi\hbar)^{-d/2}\|Q(x,y)\|_{L^2_{x,y}} = \|Q\|_{\mathcal{L}^2}.$$

Proof. The result follows from the observation that

$$W[Q](x,\xi) = (2\pi\hbar)^{-d}\mathcal{F}\left[Q\left(x+\tfrac{\cdot}{2}, x-\tfrac{\cdot}{2}\right)\right](\xi/\hbar),$$

combined with Plancherel's theorem. □

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